



## RESEARCH ARTICLE

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# Physics-Guided Deep Learning for Modeling Single-Point Wave Spectra Using Wind Inputs of Two Resolutions

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**Key Points:**

- A deep learning model for single-point directional wave spectra that is applicable for both for open ocean and coastal sea locations
- The model uses two-scale wind inputs: high-resolution for wind-seas and low-resolution for swells
- Low computational cost makes it ideal for operational wave forecasts and wave climate studies

**Supporting Information:**

Supporting Information may be found in the online version of this article.

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**Abstract** Directional Wave Spectra (DWSs) are essential for various applications such as seafaring and ocean engineering. Traditionally, DWSs are modeled with numerical wave models, which, despite their solid physical basis, are often computationally expensive. For any given point in the ocean, once the recent prior wind fields that could affect this point are known, the current DWS at this location can be largely determined. This is because local wave energy is generated by either local winds or recent prior remote winds. This fundamental physical principle can be leveraged for the statistical modeling of single-point DWSs. This study presents a deep learning approach to directly model single-point DWS using wind inputs. We utilize wind fields with two different spatial resolutions: specifically, a high-resolution, small-area, short-time local wind field for local wind-sea spectra, and a low-resolution, large-area, long-time wind field for remotely generated swell spectra. A tailored network architecture and loss function are developed to capture the relevant information from both wind scales. The performance of the proposed model was evaluated using both open ocean spectra from ERA5 and coastal spectra from numerical wave hindcasts. Results indicate that the deep learning model can effectively and efficiently simulate DWSs in both open oceans and coastal regions without relying on predefined spectral shapes, demonstrating its potential for applications in wave forecasting and wave climate studies.

**Plain Language Summary** Wave spectra describe the distribution of wave energy among different frequencies and directions, which is useful for many marine practical applications. Traditionally, wave spectra are modeled using numerical wave models, that is, numerically solving the differential equations that describe the motion of waves. Although contemporary numerical wave models perform well after years of development, their computational costs are high. The fast-developed deep learning provides an alternative to numerical wave models. In this study, we established a deep learning model that can predict the wave spectra at given locations in the ocean, by learning from data of numerical wave model. The inputs of the model are carefully selected under the guidance of physics of wave generation and propagation, which results in a model that can simulate single-point wave spectra with very low computational cost. Such a cheap but fast model is suitable for many time-sensitive and computational resource-limited applications.

## 1. Introduction

Wind-generated surface gravity waves (hereafter, waves) are the most common phenomena observed on the sea surface. Bulk wave parameters like Significant Wave Height ( $H_s$ ), Mean Wave Direction ( $D_m$ ), and Mean Wave Period ( $T_m$ ) provide basic insights into wave conditions but often lack the detail needed in complex sea states with multiple wave components. In contrast, Directional Wave Spectra (DWSs) offer a more comprehensive view by capturing wave energy distribution across different frequencies and directions, allowing for the identification of waves from various sources. This makes DWSs an important tool for ocean and coastal engineering (e.g., Holthuijsen, 2007). Accurate acquisition and modeling of DWSs are thus crucial for both scientific research and practical applications.

Numerical Wave Models (NWMs), such as WAVEWATCH III (WW3DG, 2019) and Simulating Waves Nearshore (SWAN; Booij et al., 1999), are widely used to simulate DWSs. After tens of years of development, NWMs can provide accurate wave parameters when driven by high-quality forcing and reliable boundary conditions (e.g., Alday et al., 2021; Crosby et al., 2017; Liu et al., 2021; The WISE Group, 2007). However, their computational cost remains high, especially in coastal regions, where high spatial, temporal, and spectral resolutions are required. This limits their use in time-sensitive or resource-constrained scenarios. Moreover, even

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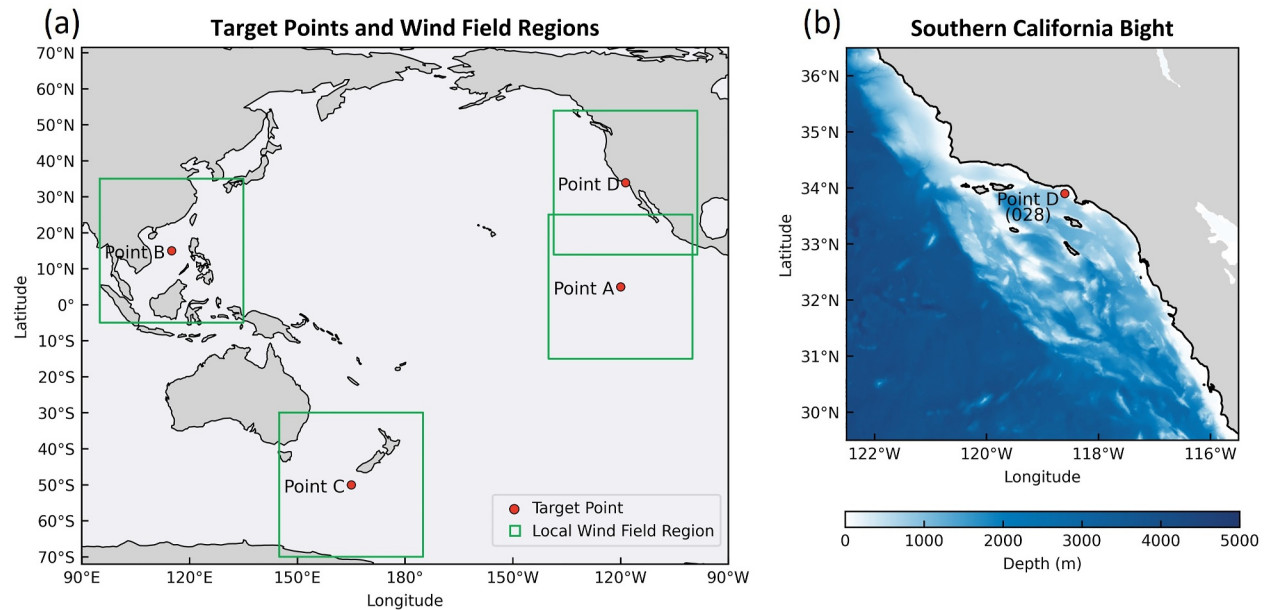
when DWSs are needed for specific locations, NWMs must model waves across entire basins, further increasing the computational load. Additionally, the performance of NWMs can be affected by incomplete physical representations and numerical effects, such as the discretization of frequency and direction during wave propagation, known as the garden sprinkler effect (Tolman, 2002), and the numerical approximation of nonlinear wave-wave interactions (Hasselmann et al., 1985).

To address these challenges, several methods for the statistical modeling of single-point DWSs have been proposed. Based on their primary input, these methods can be grouped into two categories: The first uses DWSs at open ocean boundaries to downscale them to coastal DWSs (e.g., Cagigal et al., 2024; Gao & Jiang, 2024; Ricondo et al., 2023). The underlying rationale is that waves in coastal areas are strongly influenced by bathymetry and coastal morphology, which can be considered constant over time. Consequently, once the DWSs at the open ocean boundary are known, the DWSs at a specific coastal location are largely determined. Therefore, these methods employ either hybrid modeling strategies (Cagigal et al., 2024; Ricondo et al., 2023) or deep learning techniques (Gao & Jiang, 2024) to establish the mapping relationship between coastal DWSs and open ocean DWSs. The second category of methods relies on recent prior wind or pressure fields over large areas to predict DWSs at specific locations (e.g., Espejo et al., 2014; Hegermiller, Antolinez et al., 2017; Hegermiller, Rueda, et al., 2017; Rueda et al., 2017; Song & Jiang, 2023). This approach is rooted in the physical principle that wave spectra at a given location are predominantly governed by the cumulative history of winds influencing that point, encompassing both the generation of local wind-seas and the propagation of remote swells. Thus, the second category of methods use traditional statistical tools (Espejo et al., 2014; Hegermiller, Antolinez et al., 2017; Hegermiller, Rueda, et al., 2017; Rueda et al., 2017) or deep learning (Song & Jiang, 2023) to identify the statistical relationship between DWSs at a specific point and the observed or modeled wind or pressure fields.

The rapid advancement of deep learning offers new opportunities for statistical modeling of ocean waves. Deep learning models can perform complex nonlinear regressions between inputs and outputs by training on large data sets, making them ideal when the input-output relationship is known but too intricate to express analytically. This makes deep learning a promising tool for DWS statistical modeling. As mentioned before, both of the categories of statistical modeling of single-point DWSs have corresponding deep learning versions. For the first type, Gao and Jiang (2024) applied a U-Net model to map DWSs from open ocean boundaries to nearshore locations, leveraging the fact that coastal DWS evolution is largely governed by bathymetry and coastal morphology, which remain relatively constant in a given region. For the second type, Song and Jiang (2023) used a Convolutional Neural Network to model DWSs in the open ocean, by using deep learning to capture the complex relationship between DWSs and near- and far-field winds, which is related to the generation and propagation of ocean waves in deep water that rapid surfzone evolution is not important.

While both methods show promising results, they also have limitations. Gao and Jiang's (2024) method relies on the availability of DWS data at open ocean boundaries near the target coastal point, which may not always be accessible. Song and Jiang's (2023) approach requires cumbersome pre-preprocessing steps, such as reorganizing wind data into a polar grid centered on the target point considering the land-sea distributions. Although these steps reduce the input-output size and make the model easier to train and applicable to different open-ocean locations worldwide, their method is not applicable to coastal DWSs because their direction-by-direction computation approach assumes that waves propagate along great circles. This assumption does not account for the influence of bathymetry, which play critical roles in coastal wave evolution.

To overcome the limitations of previous methods, this study introduces a deep learning model that leverages the nonlinear relationship between wind field data and DWSs at specific locations to predict single-point DWSs. Unlike Song and Jiang (2023), this model does not require reorganization of wind inputs into a polar grid, making it simpler to implement. Moreover, it is applicable for modeling DWSs in both open ocean and coastal regions. This fast and accurate data-driven surrogate is particularly useful for time-sensitive or resource-constrained scenarios where single-point DWSs are needed. Also, this study aims to demonstrate the potential of deep learning in DWSs modeling, providing profound insights and inspiration for future research. The paper is structured as follows: Section 2 introduces the study areas and data sets. Section 3 details the input, output, architecture, and training configuration of the neural network model used. Section 4 presents and analyzes the experimental results. Finally, Section 5 discusses the key findings, followed by a brief summary in Section 6.



**Figure 1.** Modeled locations and the corresponding extents of wind inputs. (a) The location of the four selected points and the spatial extent of the low-resolution-large-area wind input. The green rectangles indicate the corresponding  $40^\circ$  by  $40^\circ$  area centered on the four selected points, representing the spatial extent of the high-resolution local wind input. (b) The bathymetry (data are sourced from ETOPO, which is available at: <https://www.ncei.noaa.gov/products/etopo-global-relief-model>) of the Southern California Bight where Point D is located.

## 2. Study Locations and Data

### 2.1. Study Locations

Four locations with different wave characteristics in the Pacific Ocean, denoted as Points A, B, C, and D, are selected to demonstrate the usability of the deep learning model for DWSs (Figure 1a). These locations encompass diverse wave climates, ranging from wind-driven seas to long-period swells, from open ocean environments to coastal regions, from single-wave systems to multi-partition cases, and from small to large waves. This diversity allows for the evaluation of the model's performance under varied wave conditions.

- Point A ( $5^\circ\text{N}$ ,  $120^\circ\text{W}$ ), corresponding to Point A in Song and Jiang (2023), is located at the Pacific “swell pool” (Chen et al., 2002) and “crossing swell pool” (Jiang et al., 2017) where the westerlies-generated swell systems from both hemispheres and the trade wind-generated wave systems (can be either wind-seas or swells) often coexist at the same time (Jiang & Mu, 2019). The DWSs here are impacted by wind almost all over the Pacific Ocean, making it an idea site to examine the performance of the model under more complex, swell-dominated conditions involving the coexistence of multiple wave partitions.
- Point B ( $15^\circ\text{N}$ ,  $115^\circ\text{E}$ ), corresponding to Point B in Song and Jiang (2023), is located in the center of the South China Sea, where waves are predominantly locally generated due to the limited fetch. This point is used to represent relatively simple wind-sea-dominated conditions typical of (semi-)enclosed seas.
- Point C ( $50^\circ\text{S}$ ,  $165^\circ\text{E}$ ), consistent with Point C as defined in Song and Jiang (2023), is located in the westerlies in the Southern Ocean. This placement is specifically chosen to assess the model's efficacy in areas significantly influenced by these strong, prevailing wind systems. The wave climate at Point C is primarily wind-sea dominated, although it can also be impacted by swells originating from other sectors of the expansive Southern Ocean. By examining model performance at this location, the study aims to understand the model's capacity to accurately simulate DWS with high wave energy.
- Point D ( $33.86^\circ\text{N}$ ,  $118.64^\circ\text{W}$ ) is the location of CDIP028 coastal buoy in the Southern California Bight (Figure 1b). This location is selected for the demonstration of deep-learning downscaling of DWSs in Gao and Jiang (2024). The Southern California Bight is located along the western coast of the United States, spanning from Point Conception to the Mexican border. This area is characterized by complex and dynamic wave conditions with wave climate influenced by both local wind-sea and swells originating from the Westerlies of both the North and South Pacific (Jiang, 2020). Wave behavior here is highly variable across multiple temporal

scales—seasonal, inter-annual, and long-term—and is further modulated by complex bathymetry, coastal topography, and geographical features. Processes such as refraction, diffraction, and sheltering introduce significant small-scale variations in near-shore wave energy (e.g., Hegermiller, Rueda, et al., 2017). Therefore, this location is an optimal site for investigating the performance of the deep learning model for simulating DWSs in highly dynamic coastal environments.

## 2.2. Data

Currently, several global DWS data sets with coarse resolutions are available, including ERA5 wave data (Hersbach et al., 2020), the Center for Australian Weather and Climate Research wave hindcast (Smith et al., 2021), and the Integrated Ocean Waves for Geophysical and other Applications (IOWAGA) hindcast (Alday et al., 2021). In this study, the DWSs data from the ERA5 wind and wave data set were utilized for Points A, B, and C, which is the same input and output as Song and Jiang (2023). ERA5 is the fifth generation of the European Center for Medium Range Weather Forecasts Global Climate Atmosphere Reanalysis covering the period from 1950 to present providing hourly data on many atmospheric and wave parameters with a resolution of 31 km (Hersbach et al., 2020). The ERA5 wind data have a spatial resolution of  $0.25^\circ \times 0.25^\circ$ , while the DWS data are available at a resolution of  $0.5^\circ \times 0.5^\circ$ , with wave spectra divided into 24 directional bins at  $15^\circ$  intervals and 31 frequency bands, exponentially spaced between 0.0345 and 0.5473 Hz. The ERA5 data is available from Climate Data Store (<https://cds.climate.copernicus.eu/>) and further details about it can be found in Hersbach et al. (2020).

For Point D, because ERA5 does not provide coastal DWS data, we employed the IOWAGA hindcast data set. IOWAGA is a global wave hindcast generated using WAVEWATCH III® (WW3DG, 2019) with the physical parameterization of Ardhuin et al. (2010), forced by ERA5 global 10-m wind data, surface current fields from CMEMS-Globcurrent, and ice concentration from the IFREMER SSMI-derived daily product. Although the model does not assimilate wave observations, it shows good agreement with buoy and altimeter measurements. IOWAGA provides DWS data at more than 10,000 locations worldwide, including coastal points such as Point D. The global IOWAGA model has a resolution of  $0.5^\circ \times 0.5^\circ$ , with a two-way nested grid at a higher resolution of  $1/6^\circ$  in coastal areas, enabling dynamic downscaling for improved coastal wave simulations. The data set discretizes DWS into 24 directional sectors at  $15^\circ$  intervals, with 36 frequency bands spaced exponentially between 0.034 and 0.95 Hz, using an incremental factor of 1.1. IOWAGA data are available from the Laboratory of Physical and Space Oceanography, IFREMER (<https://www.umar-ops.fr/Donnees/Vagues>), with further details in Alday et al. (2021).

While the use of different data sets may introduce some inconsistencies in the results, it is important to note that different NWM hindcasts were selected for different locations based on data availability and suitability. Importantly, the performance of the model is not compared across locations, ensuring that any dataset-specific variations do not affect the validity of our findings. Moreover, employing different data sets offers distinct advantages. It provides a more rigorous test of the AI model's ability to surrogate different NWMs, a critical capability for AI-based wave model surrogates.

## 3. Methods

### 3.1. Model Inputs and Outputs

In this study, a deep learning approach was employed to train a model for predicting DWSs using the recent prior wind field data. The input of the model is the current and recent prior wind fields that could influence DWS at a target location, while the output is the corresponding current DWS at the target location. Different points in the ocean will have different mapping relationships with the same wind input, because their different locations. As a result, four separate models were trained for Points A, B, C, and D.

Prior research indicates that DWS at a given location can be influenced by winds in the whole basin over the past 10 days, as swells typically take about 10 days to traverse the Pacific Ocean. For instance, predicting DWS for Points A, C, and D requires ~10 days of wind data from across almost the whole Pacific as shown in Figure 1a. Without feature engineering, the input wind data for a 10-day period at a 3-hr temporal resolution and  $0.5^\circ$  spatial resolution would result in an array of shape  $2 \times 80 \times 360 \times 288$  (U and V components, 10 days  $\times$  8 fields/day, 360 longitudinal grid points, 288 latitudinal grid points, for the whole domain shown in Figure 1a).

Such a large input size for deep learning model requires a lot of graphics memory and increases the complexity of the model, which often makes the model training difficult. To address this, we applied input preprocessing techniques to reduce model complexity. First, the ERA5 wind field data in  $0.5^\circ \times 0.5^\circ$  resolution were down-loaded from Climate Data Store. Second, the  $2^\circ \times 2^\circ$  resolution data was generated by applying a  $4 \times 4$  average window to the  $0.5^\circ \times 0.5^\circ$  resolution data. Both high-resolution ( $0.5^\circ \times 0.5^\circ$ ) and low-resolution ( $2^\circ \times 2^\circ$ ) data sets were utilized as model inputs. Third, the U and V wind speed components were squared while preserving the directionality (i.e., positive or negative values), as squared wind speeds show a stronger correlation with DWSs compared to the raw U and V values (Song & Jiang, 2023). For the high-resolution wind data, only wind fields from the past 3 days within a  $40^\circ \times 40^\circ$  region centered on the target point (indicated by the green rectangle in Figure 1a) were used. The low-resolution wind data covering the entire Pacific Ocean shown in Figure 1a were also used as model input for all the four selected points.

The rationale for using such a double-scale input strategy is grounded in the dynamics of wave generation by wind. Winds near the target point have a significant and direct influence on local DWSs, particularly on the high-frequency components. In contrast, remote winds affect the target area indirectly through swell propagation. Due to frequency dispersion, angular spreading, and energy dissipation, the swell energy is considerably attenuated before reaching the target point. These processes smooth out the detailed wind patterns in the regions where the swells originate. Therefore, the finer-scale wind patterns near the target point, both spatially and temporally, are crucial for accurately modeling wind-sea DWSs. On the other hand, the wind in remote areas, which drives swell generation, can be represented at a coarser resolution.

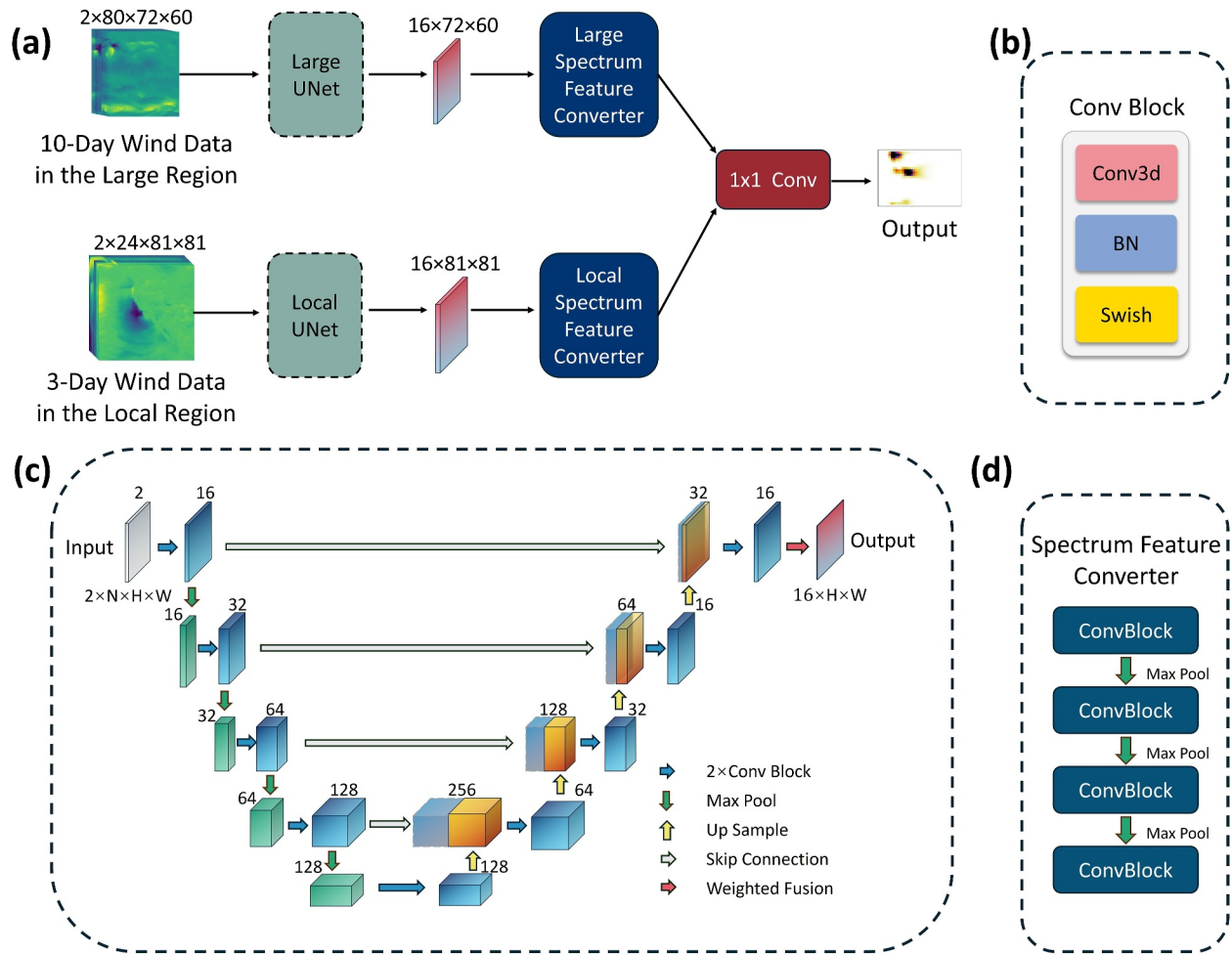
It is worth noting that different locations are often sensitive to varying temporal and spatial scales of wind forcing. For example, large-scale, coarse-resolution wind inputs may not be particularly beneficial for modeling DWSs at Point B as the South China Sea is a semi-enclose basin. Conversely, modeling DWSs along the southern coasts of Alaska may require extending the large-scale wind input to approximately 13 days, as this is the time needed for 15-s swells to travel from storm tracks in the Southern Ocean. Additionally, with trivial tuning and testing, similar or even slightly improved results could likely be achieved using slightly different model configurations while retaining the core idea of double-scale wind inputs. This includes variations in the spatial and temporal extent and resolution of the two wind fields with different resolutions. The specific details of the input data structure provided in this study are intended primarily to ensure the reproducibility of our results, rather than to assert that this particular configuration (3 days, a  $40^\circ \times 40^\circ$  region,  $0.5^\circ$  spatial resolution, and 3-hr temporal resolution for the finer-scale inputs; 10 days,  $90^\circ\text{E}$ – $90^\circ\text{W}$ ,  $2^\circ$  spatial resolution, and 3-hr temporal resolution for the coarser-scale inputs) represents an optimal configuration. We emphasize that the most critical aspect of this method is the incorporation of wind inputs at both scales, rather than the precise details of the input configuration. Meanwhile, employing a consistent input structure and model architecture across various locations simplifies the batch training and engineering application of models across different regions. As a result, the coarse-resolution wind inputs are also incorporated into the model for Point B.

Only the DWSs from NWM hindcasts are used as target output, excluding those from buoys, for reasons outlined by Song and Jiang (2023):

1. DWSs derived from buoys, reconstructed using the first five Fourier coefficients (Longuet-Higgins et al., 1963) and methods like the Maximum Entropy Method (Earle et al., 1999), are subject to significant uncertainty and tend to be noisy across spectral densities at different frequency-direction bins (e.g., Crosby et al., 2017; Jiang et al., 2022; Mooneyham et al., 2020). This makes them unsuitable as training targets for our deep learning model.
2. Buoys are sparsely distributed across the ocean, making it impossible to train a model at locations without buoy data, such as Points A, B, C. In contrast, NWMs can offer modeled DWSs at any given locations. Moreover, data-driven models require large data sets for effective training, and NWMs can theoretically generate as much data as needed. Of course, uncertainty regarding the shape of the DWS also exists for NWM outputs. However, the goal of this study is to develop a surrogate of NWMs with reduced computational costs, thus, using NWM-derived DWSs as the training target is both practical and acceptable.

### 3.2. Model Architecture

In this study, we utilized a UNet architecture as the core backbone of our model. Originally developed for medical image segmentation (Ronneberger et al., 2015), UNet has demonstrated its effectiveness in various tasks (e.g.,



**Figure 2.** The network architecture used in this study. (a) The overall model structure for predicting DWS, featuring the parallel input of high- and coarse-resolution wind data, the UNet backbone, and the multi-layer convolutional blocks before the output layer. (b) The structure of the convolutional block, which consists of two 3D convolutional layers, BatchNorm3d, and Swish activation. This structure is used throughout both the encoder and decoder. (c) The encoder-decoder design of the UNet: each colored cube represents a feature map, the number of channels shown at the top. The encoder section (left) consists of double convolutional blocks and max-pooling. The decoder section (right) reconstructs the features through upsampling, doubling the feature map size, and using skip connections. (d) The structure of the multi-layer convolutional blocks, which stack multiple convolutional layers for more extensive feature extraction and refinement.

Gao & Jiang, 2024; Sun et al., 2022). The detailed structure of the model is shown in Figure 2. As shown in Figure 2a, the model employs a two-branch architecture to process two distinct spatial resolutions wind inputs. It utilizes two separate UNet3d encoders to extract spatio-temporal features from both high-resolution, local wind fields, intended for modeling wind-seas, and low-resolution, large-area wind fields, designed for capturing swells. These encoded features are then weighted by learnable temporal parameters, enabling the model to focus on crucial time steps, followed by a summation along the time dimension to aggregate temporal information. Subsequently, dedicated spectrum feature converter transform these features into spectral representations, which are then concatenated and passed through a final  $1 \times 1$  convolutional layer to generate the predicted DWS output. This architecture effectively integrates the distinct influences of local and remote winds on wave spectra, aligning with the study's core principle of leveraging double scale wind data for accurate single-point DWS prediction. The convolutional block, shown in Figure 2b, consists of a 3D convolutional layer, a BatchNorm3d layer, and a Swish activation function. Further feature extraction is then performed using an improved UNet, whose detailed architecture is presented in Figure 2c.

The UNet architecture used in this study consists of four encoder-decoder modules. The encoder section (left side of Figure 2c) comprises double convolutional blocks, and max pooling layers, which progressively reduce the spatial dimensions of the feature maps to extract high-level features. These extracted features are passed to the

corresponding up-sampling section of the network via skip connections, allowing the model to leverage multi-scale inputs. The decoder section (right side of Figure 2c) mirrors the encoder with an equal number of decoder blocks. It restores the spatial dimensions of the feature maps by up-sampling, while incorporating skip connections to refine the outputs. The skip connections ensure that important spatial information is retained throughout the network. As the feature maps pass through the decoder, the up-sampling operations, combined with additional convolutional blocks, restore and refine the feature maps for generating accurate outputs.

Again, we provide details of the model architecture primarily to ensure the reproducibility of our results, rather than to claim that this architecture is the optimal solution for the problem. This model represents the best overall performance we achieved through numerous trivial exploratory tests. For any given input-output pair, there is almost always the potential for a better data-driven model architecture. Thus, we acknowledge that superior fitting results may be achievable using other architectures. However, the goal of this study is not to identify the most suitable model, but rather to demonstrate that using such double-scale wind inputs can effectively establish a statistical model for single-point DWSs.

### 3.3. Model Training

We used 29 years of data (1993–2021) for each buoy location, dividing it into a training set (1993–2015) and a test set (2016–2021). The training set allowed the model to “learn” the input-output relationship from recent prior data, while the independent test set was used to assess the model's performance on unseen data. Prior to feeding the wind field data into the network, we randomly shuffled the training samples to ensure robust training. At each location, spectral density values can vary drastically across different frequency and direction bins. To address this variability and improve the model's training efficiency, we applied Min-Max normalization independently to each frequency-direction bin. For each specific spectral bin, we first identified the minimum and maximum spectral density values from all spectra within the training data set for that bin across the time dimension. Then, using these bin-specific minimum and maximum values, we applied Min-Max scaling to normalize the spectral densities in all data sets for that bin to the range [0, 1]. This bin-wise normalization ensures that the spectral densities within each frequency-direction bin are scaled consistently. It is important to note that we did not normalize each individual spectrum as a whole; instead, the normalization was performed independently within each spectral bin across the training data.

The primary loss function used here is the Mean Squared Error (MSE) between the normalized spectral densities from NWM hindcasts and our model predictions, which is the same as in Gao and Jiang (2024):

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_{\text{pre}}^i - y_{\text{true}}^i)^2 \quad (1)$$

where  $m$  represents the number of samples,  $y_{\text{pre}}^i$  denotes the predicted value from our model for the  $i$ -th spectral bin, and  $y_{\text{true}}^i$  is the reference value of the normalized spectral density (from IOWAGA or ERA5) for the  $i$ -th spectral bin.

However, the standard MSE loss function showed some limitations when applied to the prediction of DWSs, which consist of large ( $30 \times 24$  matrices) and highly variable data sets. These data sets often span a wide range of values, causing the MSE to disproportionately minimizing large errors while neglecting smaller ones. This issue is particularly severe in spectra bins with a wide range of spectral densities. In this cases, the contribution of these small errors to the overall loss becomes negligible, leading the model to prioritize the reduction of larger errors. This can introduce some noise-like spurious peaks in some spectral bins. This can be observed from Figure S1 in Supporting Information S1, and also in the DWSs obtained using the method of Song and Jiang (2023).

To address this issue, we introduced an Adaptive MSE (AMSE) loss function during the fine-tuning phase of training. Specifically, after seven consecutive epochs without improvement in validation loss using the standard MSE, we switched to the AMSE loss to further optimize the model. This modified loss function adjusts the error contribution based on the magnitude of the true value, enabling the model maintain focus on both large and small values. The AMSE loss is defined as:

$$\text{AMSE} = \frac{1}{m} \sum_{i=1}^m (y_{\text{pre}}^i - y_{\text{true}}^i)^2 / 0.001 \quad (y_{\text{true}}^i < 0.001) \text{ and} \quad (2a)$$

$$\text{AMSE} = \frac{1}{m} \sum_{i=1}^m \frac{(y_{\text{pre}}^i - y_{\text{true}}^i)^2}{y_{\text{true}}^i} \quad (0.001 < y_{\text{true}}^i < 0.5) \text{ and} \quad (2b)$$

$$\text{AMSE} = \frac{1}{m} \sum_{i=1}^m (y_{\text{pre}}^i - y_{\text{true}}^i)^2 / 0.5 \quad (y_{\text{true}}^i > 0.5) \quad (2c)$$

This loss function smoothes the weighting across varying true values, ensuring that the model accounts for relative errors and absolute errors more evenly. This adaptive mechanism effectively balances the model's focus, enhancing its ability to predict DWSs without overemphasizing larger error regions at the expense of ignoring small noise-like peaks.

The maximum number of training epochs was set to 100, with an early stopping strategy implemented to enhance efficiency and avoid overfitting. The learning rate is initially set at  $10^{-4}$ , with a lower bound of  $10^{-7}$  to prevent it from becoming excessively small. If the training loss did not decrease for two consecutive epochs, the learning rate was halved. After seven epochs without improvement in validation loss, we switched to the AMSE loss and reset the learning rate to  $10^{-4}$  for fine-tuning. If the validation loss still failed to improve after another seven epochs during fine-tuning, the training was terminated. Three models were trained for the four different locations, each from scratch, using a single NVIDIA 3080Ti GPU. Training for each model took approximately 10 hr, and the inference time for predicting 1 year of DWSs at any given location was under 20 s, demonstrating the high efficiency of the model once trained.

#### 4. Results

Figure 3 presents some selected cases comparing DWSs from NWM hindcasts with estimates from the deep learning model at the four locations (full case-by-case comparisons are available as Movies S1–S4), along with comparisons of the corresponding annual mean DWSs. As shown in Figure 3, the DWSs predicted by the deep learning model align well with those from the NWM, in terms of both energy levels and spectral shapes, for both instantaneous snapshots and annual means.

At Point A, where wind-seas from the trade winds of both hemispheres and swells from the westerlies of both hemispheres often coexist, the deep learning-estimated DWSs show also good agreement with those from ERA5. The model accurately captures both the wide directional spreads of locally generated wind-sea systems by the trade winds and the narrow, sharp spectral peaks of swell systems generated by the westerlies thousands of kilometers away. This demonstrates the model's effectiveness not only in predicting wind-sea systems but also in capturing swell systems.

At Point B, the results in Figure 3 indicate that waves generated by both the summer and winter monsoons are well captured, as seen in both the selected cases and the annual mean DWSs. The shapes, locations, and energy levels of the main spectral peaks closely match the corresponding NWM results, showing the good performance of the deep learning model.

At Point C, the deep learning model demonstrates strong performance in capturing both the high-energy systems generated by local westerlies and the young swell systems originating from other storms in the Southern Ocean. This highlights the model's capability to accurately represent cases involving large wave heights and multi-modal wave conditions.

At Point D, the comparison between the 7th and 8th rows of Figure 3 again shows that the deep learning model captures the main features of coastal DWSs. Despite lacking explicit information about bathymetry and coastal morphology, the model successfully handles spectral evolution related to phenomena such as refraction, shoaling, and shadowing.

To further evaluate the performance of the deep learning model from a statistical perspective, we compared three bulk wave parameters derived from the DWSs from NWMs and deep learning models:  $H_s$ ,  $D_m$ , and  $T_m$ :

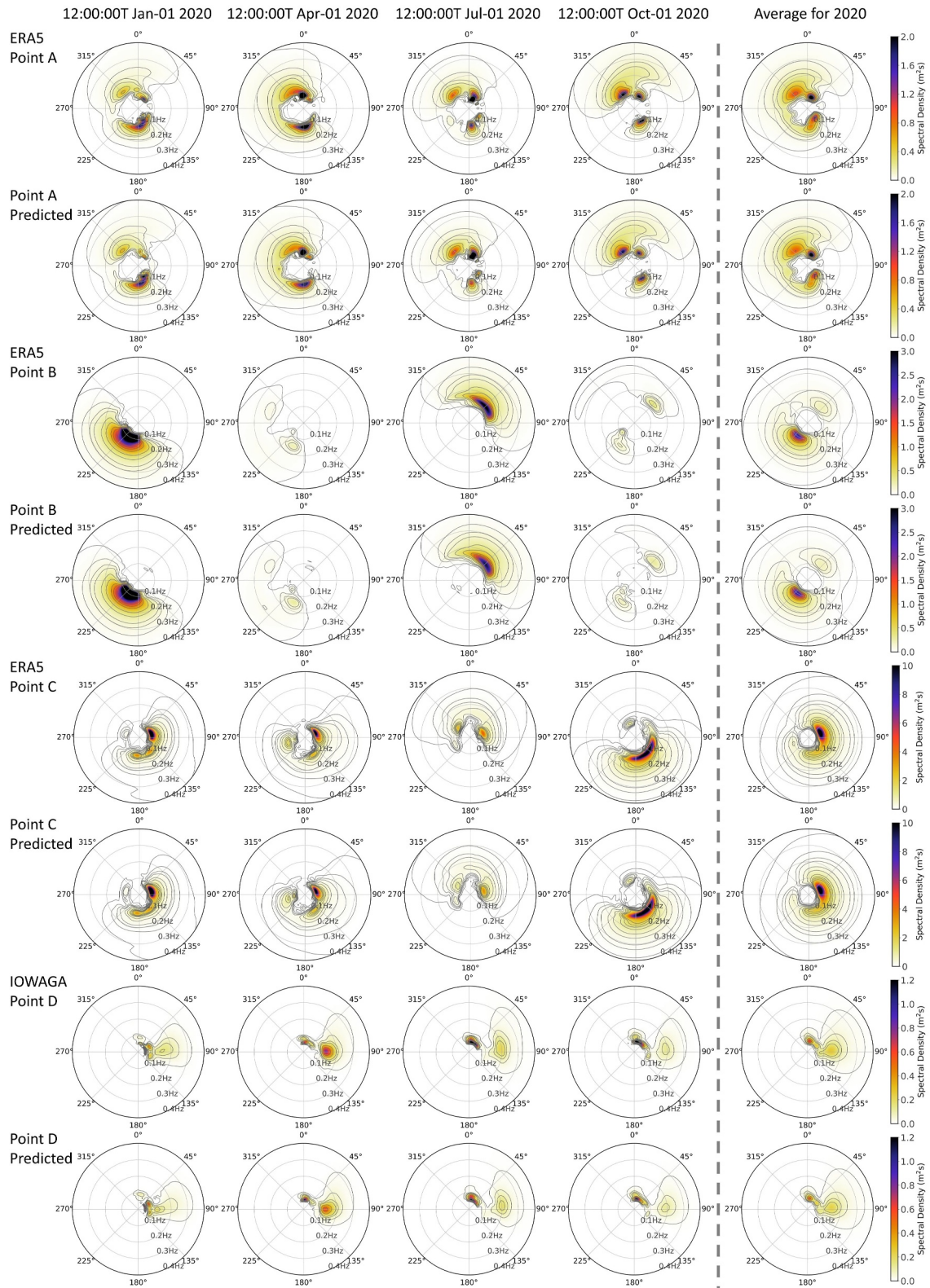


Figure 3.

$$H_s = 4.04\sqrt{m_0} \quad (3)$$

$$T_m = m_0/m_2 \quad (4)$$

$$D_m = \arctan(SF/CF) \quad (5)$$

$$m_i = \iint f^i S(f, \theta) df d\theta \quad (6)$$

$$SF = \iint \sin \theta \cdot S(f, \theta) df d\theta \quad (7)$$

$$CF = \iint \cos \theta \cdot S(f, \theta) df d\theta \quad (8)$$

where  $S(f, \theta)$  is the spectral density at frequency  $f$  and direction  $\theta$ . As noted in Section 1, these bulk parameters offer only a limited view of wave conditions and can sometimes be misleading, particularly in complex sea states with multiple peaks. Theoretically, two different DWSs can yield identical  $H_s$ ,  $D_m$ , and  $T_m$  values. However, Figure 3 and the movies in Supporting Information S1 demonstrate that the deep learning model effectively replicates the general shapes of the DWSs, validating the feasibility of using these bulk wave parameters to evaluate model performance. NWMs are also often validated using these bulk wave parameters, especially  $H_s$  (e.g., Alday et al., 2021; Liu et al., 2021), which are considered highly accurate when obtained from altimeters. Here, the bias, Root-Mean-Square-Error (RMSE), and correlation coefficient ( $R$ ) are selected as error metrics to quantitatively describe the performance of the deep learning model.

$$\text{Bias} = \frac{1}{n} \sum_{i=1}^n (y_i - x_i) \quad (9)$$

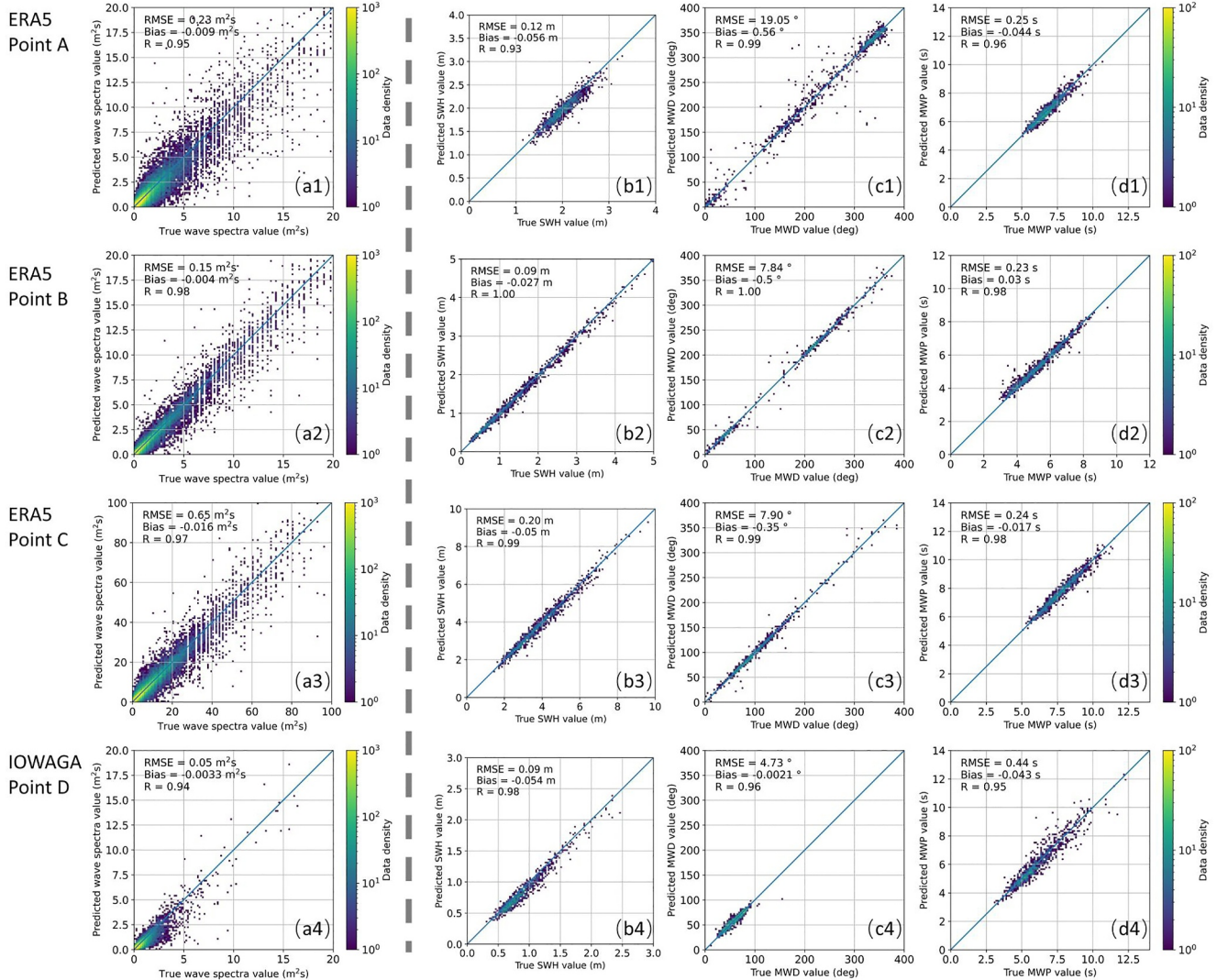
$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2} \quad (10)$$

$$R = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \quad (11)$$

where  $x$  and  $y$  denote the bulk wave parameters or spectral densities from the NWM outputs and the deep learning models, respectively, and the bars over them denote their mean values.

Figures 4a1–a4 presents scatter plots comparing the spectral density values from the deep learning model with those from NWM hindcasts across all spectral bins at the four selected locations, using the entire test data set. Although direct comparisons of spectral density values are generally not considered the best method for evaluating consistency between two sets of DWSs due to the “double penalty effect,” they still offer useful quantitative insights. The “double penalty effect” refers the phenomenon that discrepancies in the positions of narrow spectral peaks can lead to double contributions to error metrics: once for missing the correct peak and once for generating a false peak. A more detailed explanation of this effect can be found in Lledó et al. (2023). Nevertheless, these direct comparisons still serve as a valuable reference for assessing the deep learning model's performance. In these comparisons, the  $R$  values are 0.95, 0.98, 0.97, and 0.94 for Points A, B, C, and D, respectively. These values can be regarded as a good agreement considering that the  $R$  value for spectral densities

**Figure 3.** Comparison of DWSs from NWM hindcasts (1st, 3rd, 5th, and 7th rows) and deep learning model estimates (2nd, 4th, 6th, and 8th rows) at four locations in 2020. The 1st and 2nd rows are the comparisons for Point A (5°N, 120°W) in the open ocean, and 3rd and 4th rows for Point B (15°N, 115°E) in a semi-enclosed sea, and 5th and 6th rows for Point C (50°S, 165°E) in the Southern Ocean's westerlies, and 7th and 8th rows for Point D (33.86°N, 118.64°W) in the coastal sea. In each row, the four columns to the left of the dashed line represent cases at four selected time points: 12:00:00 on January 1, April 1, July 1, and October 1, 2020. The DWS to the right of the dashed line in each row represents the corresponding annual mean DWS for 2020. The directional convention used in this study follows the “marine convention,” where 0° and 90° indicate northward and eastward propagation of waves, respectively. Note that the color bars differ between the four locations.



**Figure 4.** Scatter plots between the deep learning model estimates and the NWM estimates at four selected locations (the 1st row: Point A, the 2nd row: Point B, the 3rd row: Point C, and the 4th row: Point D). The four columns correspond to comparisons of (a1, a2, a3, a4) spectral densities, (b1, b2, b3, b4)  $H_s$ , (c1, c2, c3, c4)  $D_m$ , and (d1, d2, d3, d4)  $T_m$ , respectively.

between two reconstructed DWSs from buoys located just  $\sim 13$  km apart is 0.91, as reported in Jiang et al. (2022). Moreover, the  $R$  values at Point A are 0.05 higher than those reported by Song and Jiang (2023), demonstrating the improvements achieved by our model. At Point D (Figure 4a4), although the  $R$  value is lower than the corresponding value (0.98) in Gao and Jiang (2024), it is important to note that the model inputs in the two studies are entirely different.

Figures 4b–4d show the corresponding comparisons between NWM and deep learning results for  $H_s$  (b1–b4),  $D_m$  (c1–c4), and  $T_m$  (d1–d4), respectively. The  $R$  values are 0.93, 1.00, 0.99, and 0.98, for the  $H_s$  at the four locations, and the corresponding RMSEs are 0.12, 0.09, 0.20, and 0.09 m. The relatively low  $R$  (0.93) at Point A is simply due to the narrow range of  $H_s$  values between 1.5 and 2.5 m, and the relatively large RMSE at Point C is due to the higher average  $H_s$ . For  $D_m$ , the  $R$  values exceed 0.95 at all the four points, but several outliers are found. These outliers are partly due to low  $H_s$  conditions, where minor errors (e.g., spurious peaks) in the DWS can result in significant  $D_m$  discrepancies. Another contributing factor is the “direction ambiguity” in  $D_m$  computation, which can occur when two wave systems propagate in nearly opposite directions—a common situation at Points A, B, and C. For  $T_m$ , the  $R$  values are similarly high, exceeding 0.95 at all four locations. The four bulk wave parameters show high accuracy overall, with minimal bias, low scatter, and most data points closely aligned with the  $y = x$

line. These comparisons reaffirm the deep learning model's capability to effectively model DWSs across semi-enclosed seas, open oceans, and coastal regions.

## 5. Discussions

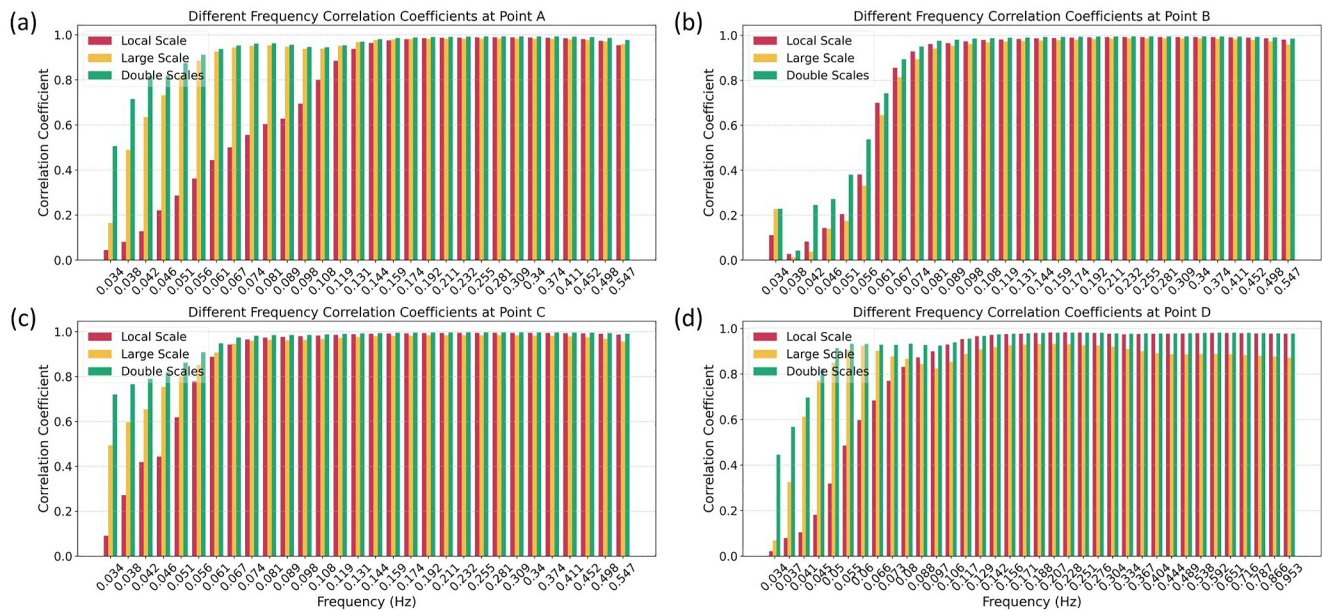
The physics behind such a deep learning DWS model using large-area recent prior wind as input has been discussed in Song and Jiang (2023) and Gao and Jiang (2024). In the open ocean, high-frequency DWSs are closely correlated with the present local wind due to the wind-wave balance during the evolution of DWSs (Jiang, 2022), making it relatively straightforward for the model to learn the quantitative relationships needed for wind-sea spectra. In contrast, the low-frequency swell DWSs involves more complex processes including generation (including processes such as wind forcing, breaking dissipation, and nonlinear wave-wave interaction) and large-distance fast propagation (including processes such as frequency dispersion, angular spreading, and non-breaking dissipation). Moreover, some numerical effect, such as the garden sprinkler effect (Tolman, 2002, referring to the numerical effect that swell propagation at coarse spectral resolution can lead to discrete swell fields in NWMs) can lead to “random” errors to the NWM output at swell frequencies. This effect contaminates the samples and hinders the model to learn low-frequency response of DWSs to recent prior remote winds. Swell spectra often appear as fast-moving narrow peaks, also exacerbating the “double penalty effect” when using an MSE-minimizing deep learning approach. All these factors collectively make data-driven swell modeling more challenging. Nevertheless, deep learning models can still capture the majority of key processes from the data, producing practical and reliable surrogate models, even for swell systems.

For the coastal location at Point D, the formation of DWSs is influenced not only by the processes affecting wind-seas and swells but also by coastal dynamics such as refraction, shoaling, and sheltering. These additional factors further increase the complexity of data-driven modeling. Fortunately, bathymetry and coastal morphology are typically considered stable, unchanging elements in wave modeling, which means that the mapping relationship between input winds and output DWSs, although complex, remains relatively fixed. The ability to uncover intricate quantitative relationships from data is a key strength of deep learning technology, making this model suitable for application not only in open oceans but also in coastal regions. However, it is important to acknowledge that NWMs have inherent limitations in shallow water regions due to factors such as simplified representations of bottom friction, wave breaking, and nonlinear wave interactions, as well as limitations in bathymetric accuracy. These constraints in the NWM hindcast data, which serve as the training target for our deep learning model, may propagate as a source of error. This should be considered when applying our AI method to coastal DWSs.

Given the rationales discussed, one might expect that a data-driven single-point DWS model could also be developed directly using the full high-resolution global wind field data from the past ~10 days, potentially yielding even better results. While this approach is theoretically feasible, using such input would require a much larger model, significantly complicating the training process. This “full-resolution” model would pose a challenge commonly seen in large-input, small-sample-size problems, where the size of the input data (i.e., the input matrix) far exceeds the sample size (i.e., the number of DWSs in the training set). In fact, when attempting to use full-resolution wind inputs, we found that the model's performance was notably inferior to the current version employing double-scale input using the same training and testing sets. Our double-scale segmentation is essentially a form of feature engineering that effectively reduces the input data volume, guided by wave physics principles. Nevertheless, we do not rule out the possibility of achieving better results with full-resolution inputs in the future, especially considering the rapid advancements in deep learning technology. Someday, more sophisticated models may emerge that can address the challenges of large input data and small sample sizes.

To further demonstrate the importance of utilizing both scales of input data, we conducted a sensitivity test by training and testing the model using either only the small-area high-resolution inputs or the large-area coarse-resolution inputs. The results are presented in Figure 5. At all four locations, it is evident that the model employing the double-scale input yields the best performance. However, the model that performs second best is different across the four points.

At Point B, where most wave energy is locally generated, the double-scale-input model and the local-scale-input model perform almost identically for frequencies above 0.1 Hz, while the large-scale-input model shows slightly inferior results due to the lack of local wind details. For frequencies below 0.1 Hz, the double-scale-input model slightly outperforms the local-scale model, likely because it better captures the swell generated more than 3 days



**Figure 5.** Correlation coefficients between deep learning-estimated and NWM-output spectral densities at different frequencies for the testing sets at three locations: (a) Point A, (b) Point B, (c) Point C, and (d) Point D. The results from different wind field inputs are represented by different colors: red for the deep learning model using only local high-resolution wind ( $40^\circ \times 40^\circ$  over 3 days), yellow for using only large-scale coarse-resolution wind across the Pacific Ocean, and green for using both inputs together.

prior. However, it is noted that there is very little wave energy at these lower frequencies at Point A, and the correlation coefficients for all four models are below 0.5 in this frequency range. Thus, while adding large-scale input data slightly improves performance, it does not significantly impact the overall results. In practice, using only local-scale high-resolution wind inputs should suffice for points within relatively small semi-enclosed seas.

At Points A, C, and D, the double-scale-input and local-scale-input models similarly perform well at frequencies above 0.15 Hz, with both slightly outperforming the large-scale-input model. However, unlike Point B, Points A, C, and D are heavily influenced by remotely generated swells, which underscores the importance of including large-scale wind input. When using only local-scale inputs, the  $R$  values drop significantly as frequency decreases—from above 0.95 at 0.13 Hz to below 0.1 at 0.034 Hz at the three locations. Adding the large-scale input dramatically improves accuracy, with  $R$  values around 0.8 for frequencies down to 0.045 Hz. The double-scale-input model also provides slightly better accuracy than the large-scale-input model at nearly all frequencies, showing that the inclusion of local wind details enhances the overall prediction. The wind-seas at Point A are primarily driven by relatively stable and homogeneous trade winds, while different wind field resolutions have minimal impact on such stable conditions. As a result, the benefit of adding high-resolution wind field for Point A is not as significant as it is for Point D. Another interesting observation is that adding high-resolution local wind data also improves low-frequency DWS predictions. This is likely due to nonlinear wave-wave interactions, which can pump high-frequency energy to lower frequencies, generating low-frequency wave energy locally. In this context, high-resolution wind fields help to better predict low-frequency DWSs as well.

In Song and Jiang (2023), some low-energy spurious peaks were observed in the deep learning-predicted DWSs at both Points A, B, and C. These spurious peaks are significantly reduced in our model due to the use of the AMSE loss function. For comparison, the results of the deep learning model trained solely with the MSE loss from Equation 1 are shown in Figure S1 of Supporting Information S1. While these models still capture the energy levels and shapes of the main peaks, spurious peaks frequently occur at frequencies around 0.07 Hz at Point A, 0.1 Hz at Points B and C, in directions where the NWM DWSs indicate no wave energy. This is because these low-energy peaks contribute very little to the overall MSE so it is difficult for the train process to eliminate these peaks. By incorporating the AMSE loss in the fine-tuning process, these spurious peaks are largely suppressed, demonstrating the necessity of using AMSE loss. Nevertheless, it is important to note that the spectral densities of

these spurious peaks are much lower than those of the “real” systems, so the integrated wave parameters from the model trained only with MSE loss still perform well (Figure S2 in Supporting Information S1).

While the proposed deep learning model demonstrates promising results in efficiently and accurately simulating single-point DWSs, several limitations must be acknowledged. First, DWSs at different locations exhibit varying responses to the same wind input due to differences in bathymetry and coastal morphology surrounding them. As a result, a model trained for a specific location cannot be directly applied to other sites, requiring a separate training process for each new point of interest. This one-model-per-location characteristic training approach may be perceived as less flexible compared to the generalized model proposed by Song and Jiang (2023), which can be applied across diverse global open ocean locations. Second, this AI model assumes constant bathymetry and coastal morphology, limiting its ability to conduct specialized experiments that NWMs can accommodate. For instance, the model cannot simulate waves under paleo-climatic conditions, where historical land-sea distributions may have differed significantly. Third, the current version of the AI model does not incorporate sea ice or ocean currents, which may result in significant errors in polar regions and areas with strong currents. Finally, the accuracy of the deep learning approach is intrinsically constrained by the quality and precision of the NWM data used for training.

However, our model offers several key advantages. First, its performance is slightly better than Song and Jiang (2023) when trained on a single point and significantly better than their generic model. More importantly, our model is applicable not only in the open ocean but also in coastal seas, where the method of Song and Jiang (2023) is not applicable. Additionally, from a practical standpoint, our model is much simpler to implement, as it avoids the complex pre-processing of wind fields. In Song and Jiang (2023), wind inputs must be re-organized into a polar grid centered on the target point, and wind vectors need to be re-projected relative to the target location, accounting for land-sea distributions. In contrast, our model retains the wind inputs in their original rectangular grid, with the only pre-processing step being the selection of the relevant region, making it far more user-friendly.

## 6. Summary

In this study, we present a deep learning approach to model single-point DWS based on wind inputs over a relatively large spatiotemporal extent. This approach is grounded in the principle that wave energy is generated by either current local winds or recent prior remote winds. To facilitate training and enhance model performance, we employ a double-scale wind input strategy: a high-resolution, small-area, short-time local wind field simulates local wind-sea DWSs, while a low-resolution, large-area, long-time wind field accounts for remotely generated swell DWSs. The performance of this deep learning model was evaluated using DWSs from four locations with distinct wave characteristics: Point A, located in the open ocean swell pool where wind-sea and swell systems often coexist; Point B, situated in a semi-enclosed sea predominantly influenced by wind-seas; Point C, located in the Southern Ocean's westerlies; and Point D, found in a coastal area characterized by both wind-seas and swells. The results indicate that the deep learning model effectively and efficiently models DWS at any given location with low computational costs.

This model can serve as an alternative to NWMs in single-point DWS modeling, particularly in time-sensitive or resource-limited scenarios. Potential applications of this method include modeling DWSs in underdeveloped regions, generating ensemble forecasts of catastrophic waves during emergencies, and conducting wave climate studies. For wave forecasting, the well-trained AI model can be driven by an ensemble of wind field predictions to produce efficient ensemble wave forecasts. Additionally, it is worth noting that this AI model does not require any “initial conditions,” enabling rapid updates to the forecast results as soon as the assimilation system provides updated wind information. For wave climate studies, an increasing number of investigations are focusing on wave climate at the spectral level (e.g., Echevarria et al., 2019; Espejo et al., 2014; Jiang & Mu, 2019; Lobeto et al., 2022). However, ensemble modeling of waves under different climate scenarios for long-term projections can be computationally expensive using NWMs, especially when DWSs are required only at specific locations. Once the deep learning model is trained for a given location, which takes several hours on a personal computer, it can model tens of thousands of DWSs at that location within minutes, facilitating efficient predictions under various future scenarios.

Future work may focus on further improving the performance of this deep learning model. First, even with the same input and target output, the model can be optimized using more advanced deep learning frameworks.

Second, when employing a more effective deep learning architecture, additional inputs can be considered, such as the aforementioned “full-resolution” wind data and other physically relevant forces, including ocean currents.

## Data Availability Statement

All data used in this study are publicly accessible online. The DWSs data from IOWAGA can be downloaded from the official website of The Laboratory of Physical and Space Oceanography, IFREMER (<https://www.umr-lops.fr/Donnees/Vagues>). The ERA5 wind fields and DWSs data are available from the ECMWF Climate Data Store (<https://cds.climate.copernicus.eu/>). The deep learning models are implemented using PyTorch (<https://pytorch.org/>). The source code and associated documents are available in the Zenodo (<https://doi.org/10.5281/zenodo.14812406>) repository and the GitHub repository (<https://github.com/oaen/wind2dws>).

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